Phase 4 Cross-modal Similarity Summary

Phase 4: Cross-modal similarity summary

Question:

Do embedding-space neighbors match SAR-space neighbors?

Overall k-NN overlap:

- k=5: mean overlap=0.020, median overlap=0.000
- k=10: mean overlap=0.032, median overlap=0.000
- k=20: mean overlap=0.053, median overlap=0.000

At k=10, the observed overlap (0.032) is above the random-match baseline (~0.003), but the median query still has zero overlap.

Overall pooled distance correlation: Pearson r=0.195, Spearman rho=0.151

Within-region Pearson correlations range from 0.275 (iowa_ag) to 0.689 (california_coast).

This suggests pooled cross-region comparisons dilute a stronger within-region relationship.

Highest land-use overlap at k=10:

```
water: mean overlap=0.098
snow_and_ice: mean overlap=0.043
flooded_vegetation: mean overlap=0.036
```

Lowest land-use overlap at k=10:

```
crops: mean overlap=0.014
shrub_and_scrub: mean overlap=0.015
grass: mean overlap=0.016
```

Representative Sentinel-2 RGB chips are included for the query pixel, the top embedding-space neighbor, and the top SAR-space neighbor.

Phase 4 Overall k-NN Overlap

k	n_queries	mean_overlap	median_overlap	p10_overlap	p90_overlap
5	2880	0.020	0.000	0.000	0.000
10	2880	0.032	0.000	0.000	0.100
20	2880	0.053	0.000	0.000	0.150

Overlap is the fraction of embedding-space k-nearest neighbors that also appear in SAR-space k-nearest neighbors.

Phase 4 Land-use Overlap at k=10

dw_label_name	n_queries	mean_overlap	median_overlap	p10_overlap	p90_overlap
water	320	0.098	0.100	0.000	0.300
snow_and_ice	320	0.043	0.000	0.000	0.100
flooded_vegetation	320	0.036	0.000	0.000	0.100
trees	320	0.024	0.000	0.000	0.100
bare	320	0.022	0.000	0.000	0.100
built	320	0.020	0.000	0.000	0.100
grass	320	0.016	0.000	0.000	0.100
shrub_and_scrub	320	0.015	0.000	0.000	0.100
crops	320	0.014	0.000	0.000	0.100

The land-use table shows where cross-modal neighbor agreement is strongest or weakest.

Phase 4 Region Overlap at k=10

region	n_queries	mean_overlap	median_overlap	p10_overlap	p90_overlap
california_coast	720	0.041	0.000	0.000	0.100
sf_bay_urban	720	0.040	0.000	0.000	0.100
amazon_forest	720	0.030	0.000	0.000	0.100
iowa_ag	720	0.018	0.000	0.000	0.100

Regional overlap helps assess whether similarity alignment is geography-dependent.

Phase 4 Distance Correlation

scope	group	n_pairs	pearson_r	spearman_rho
overall	all	200000	0.195	0.151
region	sf_bay_urban	258840	0.508	0.519
region	iowa_ag	258840	0.275	0.272
region	amazon_forest	258840	0.530	0.506
region	california_coast	258840	0.689	0.676
dw_label_name	water	51040	0.574	0.585
dw_label_name	trees	51040	0.172	0.195
dw_label_name	grass	51040	0.360	0.331
dw_label_name	flooded_vegetation	51040	0.372	0.355
dw_label_name	crops	51040	0.186	0.118
dw_label_name	shrub_and_scrub	51040	0.267	0.268
dw_label_name	built	51040	0.115	0.092
dw_label_name	bare	51040	0.145	0.130
dw_label_name	snow_and_ice	51040	0.220	0.217

Pairwise distance correlation compares embedding cosine distance to SAR Euclidean distance.

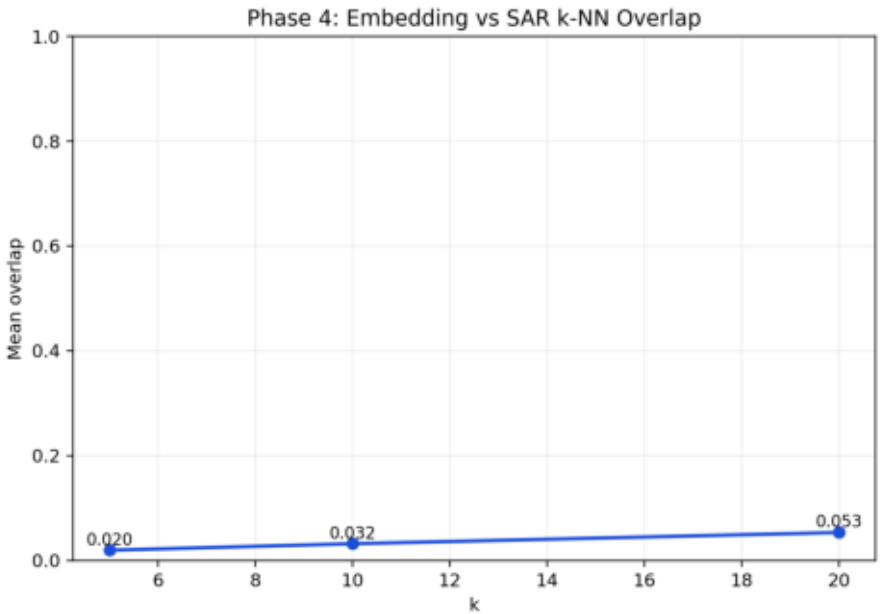
Phase 4 Representative Queries

reason	region	dw_lab	overlap	embedding_neighbor_labels	sar_neighbor_labels
highest_ove	sf_bay_	water	0.700	water@sf_bay_urban, water@sf_bay_urban, water@sf_bay_urban, water@sf_bay_urban, water@sf_bay_urban	water@sf_bay_urban, water@sf_bay_urban, water@sf_bay_urban, water@california_coast, water@sf_bay_urban
lowest_overl	sf_bay_	water	0.000	water@sf_bay_urban, water@sf_bay_urban, water@sf_bay_urban, water@sf_bay_urban, water@sf_bay_urban	bare@sf_bay_urban, water@iowa_ag, water@iowa_ag, water@sf_bay_urban, water@sf_bay_urban
water_repre	sf_bay_	water	0.100	water@sf_bay_urban, water@sf_bay_urban, water@sf_bay_urban, water@sf_bay_urban, water@sf_bay_urban	water@sf_bay_urban, water@amazon_forest, water@california_coast, water@amazon_forest, water@amazon_forest
crops_repres	sf_bay_	crops	0.000	crops@sf_bay_urban, bare@sf_bay_urban, bare@sf_bay_urban, crops@sf_bay_urban, bare@sf_bay_urban	crops@california_coast, bare@iowa_ag, crops@iowa_ag, flooded_vegetation@iowa_ag, shrub_and_scrub@sf_bay_urban

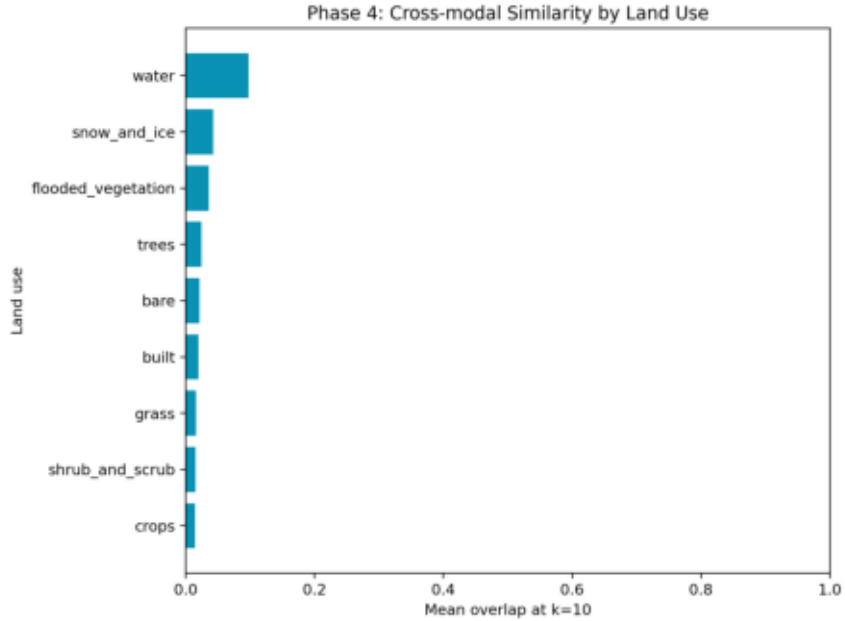
Representative queries provide a qualitative check on where neighbor sets align or diverge.

Phase 4 Visual Diagnostics

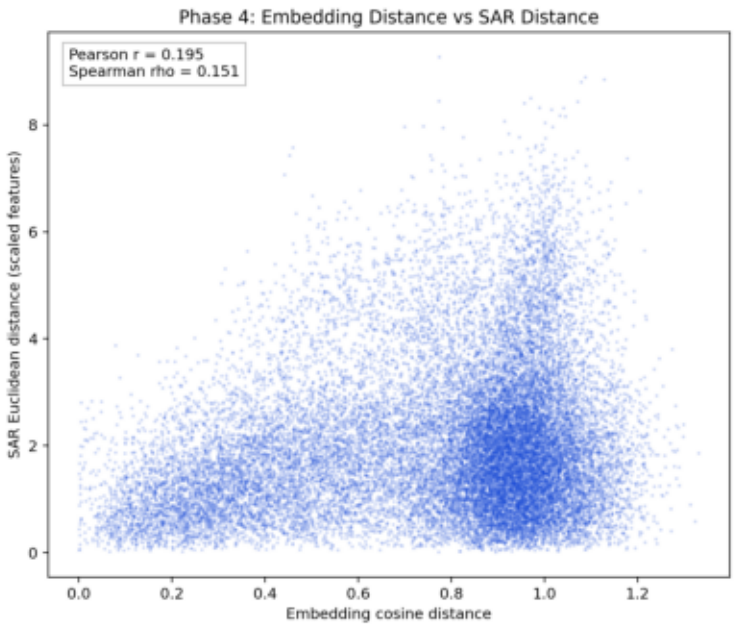
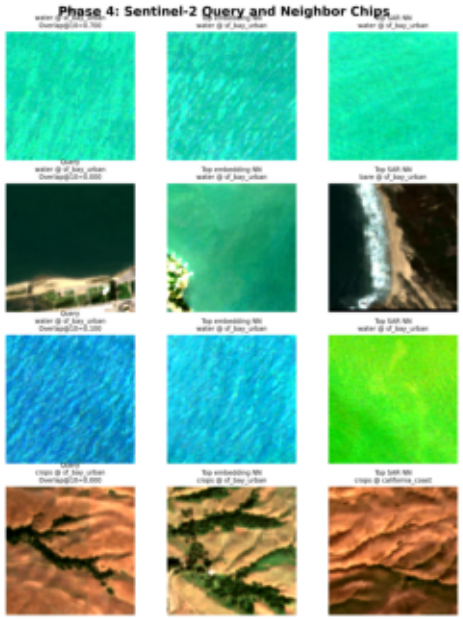
phase4 knn overlap by k



phase4 knn overlap by land use k10



phase4 embedding vs sar distance



phase4 sentinel2 neighbor chips